Pure Mathematics Year 2

Chapter 11: Integration

Standard functions, reverse chain rule, substitution, parts, partial fractions, differential equations and the trapezium rule.

Chapter Overview — Integration Methods 2

In this chapter you will integrate a greater variety of expressions and solve differential equations.

Standard results

Know the antiderivatives of standard functions.

$$\int \sec^2 x \, dx = \tan x + C.$$

Substitution

Introduce a new variable to make an integral easier.

$$\int x\sqrt{2x+5}\,dx$$

$$u = 2x+5, \quad x = \frac{u-5}{2}, \quad dx = \frac{1}{2}\,du.$$

$$\int x\sqrt{2x+5}\,dx = \frac{1}{4}\int (u-5)\sqrt{u}\,du.$$

Reverse chain rule

Think about what differentiates to give the expression.

$$\int \cos 4x \, dx = \frac{1}{4}\sin 4x + C.$$

$$\int \sin^3 x \cos x \, dx = \frac{1}{4}\sin^4 x + C.$$

Integration by parts

Integrate a product using the product rule in reverse.

$$\int x \cos x \, dx$$

$$u = x, \quad \frac{dv}{dx} = \cos x, \quad \frac{du}{dx} = 1, \quad v = \sin x.$$

$$\int x \cos x \, dx = x \sin x - \int \sin x \, dx$$

$$= x \sin x + \cos x + C.$$

Integrating partial fractions

Split a rational expression into partial fractions so that each term is easier to integrate.

$$\int \frac{3x+5}{(x+1)(x+2)}\,dx$$

Solving differential equations

Find one variable in terms of another without derivatives remaining. For $V > 0$:

$$\frac{dV}{dt} = -kV$$
$$\int \frac{1}{V}\,dV = \int -k\,dt$$
$$\ln V = -kt + C$$
$$V = e^{-kt+C} = Ae^{-kt}.$$

Approximating areas: the trapezium rule

Instead of integrating, split the area under the graph into trapeziums and add their areas to approximate the total.

Integrating Standard Functions 4

$\int x^n\,dx =$

$\frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$

$\int e^x\,dx =$

$e^x + C$

$\int \frac{1}{x}\,dx =$

$\ln|x| + C$

The absolute value allows negative x; $x = 0$ is excluded.

$\int \cos x\,dx =$

$\sin x + C$

Picture sine above cosine: down is differentiating; up is integrating. Change the sign when travelling the other way.

$\int \sin x\,dx =$

$-\cos x + C$

$\int \sec^2 x\,dx =$

$\tan x + C$

Memorise $\int \sec^2 x\,dx = \tan x + C$.

$\int \operatorname{cosec} x \cot x\,dx =$

$-\operatorname{cosec} x + C$

$\int \operatorname{cosec}^2 x\,dx =$

$-\cot x + C$

$\int \sec x \tan x\,dx =$

$\sec x + C$

Quickfire Questions 5

$\int \sec x \tan x \, dx =$

1

$\int \sin x \, dx =$

2

$\int \operatorname{cosec}^2 x \, dx =$

3

$\int -\cos x \, dx =$

4

Quickfire Questions 6

$\int \sec^2 x \, dx =$

1

$\int \operatorname{cosec} x \cot x \, dx =$

2

$\int 1/x \, dx =$

3

$\int -\sin x \, dx =$

4

Quickfire Questions — Without Looking Back 7

$\int \operatorname{cosec}^2 x \, dx =$

1

$\int \sin x \, dx =$

2

$\int \sec x \tan x \, dx =$

3

$\int \cos x \, dx =$

4

Test Your Understanding — Standard Integrals 8

1. Find $\int (2\cos x + 3/x - \sqrt{x}) \, dx$.

1

2. Find $\int \frac{\cos x}{\sin^2 x} \, dx$.

Hint: Which reciprocal trigonometric functions does this simplify to?

2

3. Given $\int_a^{3a} \frac{2x+1}{x} \, dx = \ln 12$, find the exact value of a.

3

Note

You may split a numerator:

$$\frac{x+1}{x} = \frac{x}{x} + \frac{1}{x} = 1 + \frac{1}{x}.$$

You cannot split a denominator:

$$\frac{x}{x+1} \neq \frac{x}{x} + \frac{x}{1}.$$

4. Given that $a > 0$ and $\int_a^{2a} \frac{3x-1}{x}\,dx = 6 + \ln(\frac{1}{2})$, find the exact value of a.

4

5. Given that $a > 0$ and $\int_{\ln 1}^{\ln a} (e^x + e^{-x})\,dx = 48/7$, find the exact value of a.

5

7. $f(x) = \frac{1}{8}x^{3/2} - 4/x,\ x > 0.$

a. Solve $f(x) = 0$.

b. Find $\int f(x)\,dx$.

c. Evaluate $\int_1^4 f(x)\,dx$, giving your answer as $p + q\ln r$, where p, q, r are rational numbers.

7a

7b

7c

Integrating a Linear Inner Function 11

$\dfrac{d}{dx}\sin(3x+1) =$

$3\cos(3x+1).$

Therefore $\int \cos(3x+1)\,dx =$

$\frac{1}{3}\sin(3x+1) + C.$

For a linear inner function $ax+b$, integrate as before and divide by a, where $a \neq 0$:

$$\int f'(ax+b)\,dx = \frac{1}{a}f(ax+b) + C.$$

1. $\int e^{3x}\,dx$

1

2. $\int \frac{1}{5x+2}\,dx$

2

3. $\int 2\sec^2(3x-2)\,dx$

3

4. $\int (3x+4)^3\,dx$

4

5. $\int \sin(1-5x)\,dx$

5

6. $\int \frac{1}{3(4x-2)^2}\,dx$

6

7. $\int (10x+11)^{12}\,dx$

7

Tip

For $\int (ax+b)^n\,dx$, divide by both $n+1$ and a ($n \neq -1$, $a \neq 0$).

Check Your Understanding — Linear Inner Functions 12

$\int e^{3x+1}\,dx =$

1

$\int \frac{1}{1-2x}\,dx =$

2

$\int (4-3x)^5\,dx =$

3

$\int \sec(3x)\tan(3x)\,dx =$

4

Find the indefinite integral of each expression with respect to x.

1a. $\sin(2x+1)$

1a

1c. $4e^{x+5}$

1c

1e. $\operatorname{cosec}^2(3x)$

1e

1f. $\sec 4x \tan 4x$

1f

1g. $3\sin(\frac{1}{2}x+1)$

1g

1h. $\operatorname{cosec} 2x \cot 2x$

1h

2a. $e^{2x} - \frac{1}{2}\sin(2x-1)$

2a

2b. $(e^x+1)^2$

2b

2c. $\sec^2(2x)(1+\sin 2x)$

2c

2d. $\frac{3-2\cos(\frac{1}{2}x)}{\sin^2(\frac{1}{2}x)}$

2d

2e. $e^{3-x} + \sin(3-x) + \cos(3-x)$

2e

3a. $\frac{1}{2x+1}$

3a

3b. $\frac{1}{(2x+1)^2}$

3b

3c. $(2x+1)^2$

3c

3d. $\frac{3}{4x-1}$

3d

3f. $\frac{3}{(1-4x)^2}$

3f

3h. $\frac{3}{(1-2x)^3}$

3h

3j. $\frac{5}{3-2x}$

3j

4a. $3\sin(2x+1) + \frac{4}{2x+1}$

4a

4c. $\frac{1}{\sin^2 2x} + \frac{1}{1+2x} + \frac{1}{(1+2x)^2}$

4c

6. Given $\int_3^b (2x-6)^2\,dx = 36$, find b.

6

7. Given $\int_{e^2}^{e^8} \frac{1}{kx}\,dx = \frac{1}{4}$, find k.

7

8. Given $\int_{\pi/(4k)}^{\pi/(3k)} (1 - \pi \sin kx)\,dx = \pi(7 - 6\sqrt{2})$, find the exact value of k.

8

Challenge

Given $\int_{5}^{11} \frac{1}{ax+b}\,dx = \frac{1}{a}\ln(41/17)$, where a, b are integers and $0 < a < 10$, find two different pairs of values of a, b.

Working

Case 1

Case 2

Integrating Using Trigonometric Identities 17

Some expressions, including $\sin^2 x$ and $\sin x \cos x$, need a trigonometric identity before integration.

Find $\int \sin^2 x \, dx$.

Which identity contains $\sin^2 x$?

Sine squared

Find $\int \cos^2 x \, dx$.

Cosine squared

Find $\int \sin 3x \cos 3x \, dx$.

Product

Find $\int \tan^2 x \, dx$.

Tangent squared

Check Your Understanding — Trigonometric Identities 18

Find $\int (\sec x + \tan x)^2 \, dx$.

1

Find $\int_{\pi/6}^{\pi/3} \sin^2(3x) \, dx$.

2

3. Show that $\int_{\pi/4}^{\pi/2} \sin^2 x \, dx = \frac{2+\pi}{8}$.

3

5a. Expand $\sin(3x + 2x)$ and $\sin(3x - 2x)$, using the compound-angle formulae. Hence show that

$$\sin 5x + \sin x = 2 \sin 3x \cos 2x.$$

5a

5b. Hence find $\int \sin 3x \cos 2x \, dx$.

5b

6. $f(x) = 5 \sin^2 x + 7 \cos^2 x$.

a. Show that $f(x) = \cos 2x + 6$.

6a

6b. Find $\int_0^{\pi/4} f(x) \, dx$.

6b

7a. Show that

$$\cos^4 x = \tfrac{1}{8} \cos 4x + \tfrac{1}{2} \cos 2x + \tfrac{3}{8}.$$

7a

7b. Hence find $\int \cos^4 x \, dx$.

7b

Reverse Chain Rule — Consider Then Scale 21

Complicated expressions can look like the result of applying the chain rule. Use “consider then scale”.

1. Consider an expression whose derivative will look similar.
2. Differentiate and adjust for any scale difference.

$\int x(x^2+5)^3\,dx$

The first x looks as though it arose from differentiating x^2 inside the brackets.

1

$\int \cos x \sin^2 x\,dx$

The $\cos x$ probably arose from differentiating $\sin x$.

2

$\int \frac{2x}{x^2+1}\,dx$

The $2x$ probably arose from differentiating x^2.

3

Reverse Chain Rule — Common Patterns 22

Integration by inspection / reverse chain rule

Consider an expression that would differentiate to the expression given. Then scale appropriately.

Common patterns:

$\int k\frac{f'(x)}{f(x)}\,dx$: try $\ln|f(x)|$.

$\int kf'(x)[f(x)]^n\,dx$: try $[f(x)]^{n+1}$ $(n \neq -1)$.

If the bottom of a fraction differentiates to give the top, apart from a constant scale factor, try the logarithm of the absolute value of the bottom.

$\int \frac{x^2}{x^3+1}\,dx$

1

$\int xe^{x^2+1}\,dx$

2

Quickfire — Reverse Chain Rule 23

In your head!

$\int \frac{4x^3}{x^4-1}\,dx =$

1

$\int \frac{\cos x}{\sin x+2}\,dx =$

2

$\int \cos x\, e^{\sin x}\,dx =$

3

$\int \cos x(\sin x - 5)^7\,dx =$

4

$\int x^2(x^3+5)^7\,dx =$

5

Not in your head...

$\int \frac{x}{(x^2+5)^3}\,dx$

6

Tip

If the whole denominator is raised to a power, do not apply the logarithm pattern directly. Rewrite the expression as a product, such as $x(x^2+5)^{-3}$, then use the power pattern.

Powers of Sine and Secant 24

When we differentiate $\sin^5 x$, the power decreases:

$$\frac{d}{dx}(\sin^5 x) = \frac{d}{dx}[(\sin x)^5] = 5\sin^4 x\cos x.$$

However, when we differentiate $\sec^5 x$:

$$\frac{d}{dx}[(\sec x)^5] = 5\sec^4 x \times \sec x\tan x = 5\sec^5 x\tan x.$$

The power of $\sec x$ does not decrease. Keep this in mind when integrating.

Find $\int \sec^4 x\tan x\,dx$.

Integral

$\int \sin x(\cos x + 1)^5 \, dx$

1

$\int \dfrac{\operatorname{cosec}^2 x}{(2 + \cot x)^3} \, dx$

2

$\int x(x^2 + 2)^3 \, dx$

4

$\int \dfrac{\sec^2 2x}{\tan 2x + 1} \, dx$

3

$\int 5 \tan x \sec^2 x \, dx$

5

Exercise 11D — Pages 302–303 26

4. Given that $\int_0^k kx^2 e^{x^3} \, dx = \frac{2}{3}(e^8 - 1)$, find k.

4

5. Given that $\int_0^\theta 4 \sin 2x \cos^4 2x \, dx = \frac{4}{5}$, where $0 < \theta < \pi$, find the exact value of θ.

5

6a. By writing $\cot x = \cos x / \sin x$, find $\int \cot x \, dx$.

6a

6b. Show that $\int \tan x \, dx = \ln |\sec x| + C$.

6b

Extension — STEP I 2013, Q4

i. Show that, for $n > 0$,

$$\int_0^{\pi/4} \tan^n x \sec^2 x \, dx = \frac{1}{n+1}$$

and $$\int_0^{\pi/4} \sec^n x \tan x \, dx = \frac{(\sqrt{2})^n - 1}{n}.$$

ii. Evaluate

$$\int_0^{\pi/4} x \sec^4 x \tan x \, dx$$

and $$\int_0^{\pi/4} x^2 \sec^2 x \tan x \, dx.$$

Extension ii

Integration by Substitution 28

A substitution can turn a complicated integral into an equivalent, simpler integral. The following is not a direct reverse-chain-rule pattern.

Use $u = 2x + 5$ to find $\int x\sqrt{2x+5} \, dx$.

Remove every reference to x, replacing it with u. Work out expressions for x and dx.

Step 1

Work out x and dx, or a useful variant.

1

Step 2

Substitute into the integral.

2

Step 3

Integrate the simplified expression.

3

Step 4

Write the answer in terms of x.

4

Choosing a Substitution 29

Edexcel questions usually give the substitution. Other examination boards and STEP may not.
There is no fixed rule: look for expressions inside roots, powers or the denominator of a fraction.
Suggest a sensible substitution for each integral.

$\int \cos x\sqrt{1+\sin x}\,dx$

1

$\int 6xe^{x^2}\,dx$

2

$\int \frac{xe^x}{1+x}\,dx$

3

$\int e^{(1-x)/(1+x)}\,dx$

4

Examples 3 and 4 practise choosing a substitution only. Their antiderivatives are not elementary, so these substitutions do not complete an integration using this chapter's methods.

Another Substitution Example 30

Use $u = \sin x + 1$ to find $\int \cos x \sin x(1+\sin x)^3\,dx$.

Step 1

Work out x and dx, or useful variants.

1

Step 2

Substitute into the integral.

2

Step 3

Integrate.

3

Step 4

Write the answer in terms of x.

4

Substitution with Implicit Differentiation 31

When a square root is involved, a substitution of the form $u^2 = \cdots$ can keep the algebra tidy.
Use $u^2 = 2x + 5$, taking $u = \sqrt{2x+5} \geq 0$, to find $\int x\sqrt{2x+5}\,dx$.

Step 1

Find x and dx.

1

Step 2

Substitute.

2

Step 3

Integrate.

3

Step 4

Return to x.

4

This avoids the fractional powers that arose when using $u = 2x + 5$.

Test Your Understanding — Substitution 32

Edexcel C4 January 2012, Q6c

Using $u = 1 + \cos x$, or otherwise, show that

$$\int \frac{2\sin 2x}{1+\cos x}\,dx = 4\ln(1+\cos x) - 4\cos x + k,$$

where k is a constant.

Hint: use the double-angle formula first.

Working

Definite Integration — Change the Limits 33

Calculate $\int_0^{\pi/2} \cos x \sqrt{1+\sin x}\, dx$.

Choose a substitution.

1

2

Find the new limits when $x = \pi/2$ and when $x = 0$.

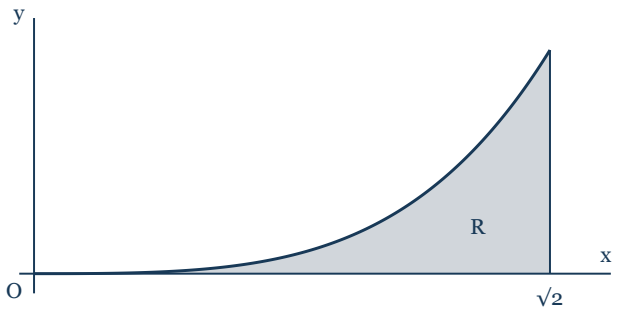
3

4

Test Your Understanding — Area Substitution 34

Edexcel C4 June 2011, Q4c

The curve is $y = x^3 \ln(x^2+2)$, $x \geq 0$.



The finite region R is bounded by the curve, the x-axis and $x = \sqrt{2}$.

Use $u = x^2 + 2$ to show that its area is

$$\frac{1}{2}\int_2^4 (u-2)\ln u\, du.$$

Working

5. Using $u^2 = 4x + 1$, or otherwise, find the exact value of $\int_6^{20} \frac{8x}{\sqrt{4x+1}}\,dx$.

5

6. Use $u^2 = e^x - 2$ to show that

$$\int_{\ln 3}^{\ln 4} \frac{e^{4x}}{e^x - 2}\,dx = \frac{a}{b} + c\ln d,$$

where a, b, c, d are integers to be found.

6

7. Prove that $-\int \frac{1}{\sqrt{1-x^2}}\,dx = \arccos x + C$.

7

8. Use $u = \cos x$ to show that

$$\int_0^{\pi/3} \sin^3 x \cos^2 x\,dx = \frac{47}{480}.$$

8

9. Using a suitable trigonometric substitution for x, find

$$\int_{1/2}^{\sqrt{3}/2} x^2\sqrt{1-x^2}\,dx.$$

9

Challenge

Using a substitution of the form $x = k\sin u$, show that

$$\int \frac{1}{x^2\sqrt{9-x^2}}\,dx = -\frac{\sqrt{9-x^2}}{9x} + C.$$

Challenge

Integration by Parts 38

How can we find $\int x\cos x\,dx$?

Just as the product rule differentiates a product of two expressions, integration by parts can often integrate a product.

Integration by parts

$$\int u\frac{dv}{dx}\,dx = uv - \int v\frac{du}{dx}\,dx.$$

Proof (not needed for the examination)

The product rule gives

$$\frac{d}{dx}(uv) = v\frac{du}{dx} + u\frac{dv}{dx}.$$

Both terms on the right are products. We can let $u\frac{dv}{dx}$ represent a product such as $x\cos x$.

Rearranging:

$$u\frac{dv}{dx} = \frac{d}{dx}(uv) - v\frac{du}{dx}.$$

Integrating both sides with respect to x gives the formula above.

Integration by Parts — Choosing the Factors 39

Find $\int x \cos x \, dx$.

Step 1

Decide which factor is u and which is dv/dx. Choose u so that differentiating makes it simpler, and choose dv/dx so that it is easy to integrate. For the standard products here, choose the x^n term as u, unless the other factor is $\ln x$.

1

Step 2

Find du/dx and v.

2

Step 3

Use the formula.

$$\int u\frac{dv}{dx}\,dx = uv - \int v\frac{du}{dx}\,dx.$$

3

Memory aid: "uv minus the integral of the two new expressions multiplied together".

Another Integration-by-Parts Example 40

Find $\int x^2 \ln x \, dx$, for $x > 0$.

Step 1

Choose u and dv/dx.

Step 2

Find du/dx and v.

Step 3

Use the integration-by-parts formula.

Working

Integration by Parts Twice 41

Find $\int x^2 e^x \, dx$.

Tip: write the working for the second integral separately, then substitute its result into the original calculation.

Working

Test Your Understanding — Parts Twice 42

Find $\int x^2 \sin x \, dx$.

Answer

Integrating Logarithms and Definite Integrals 43

Find $\int \ln x \, dx$, for $x > 0$.

Indefinite

Find $\int_1^2 \ln x \, dx$, leaving the answer in terms of natural logarithms.

Definite

General definite-integral formula

$$\int_a^b u\frac{dv}{dx}\,dx = [uv]_a^b - \int_a^b v\frac{du}{dx}\,dx.$$

Test Your Understanding — Definite Integration by Parts 44

Find $\int_0^{\pi/2} x \sin x \, dx$.

Working

When the Original Integral Returns

Find $\int e^x \sin x \, dx$.

Either factor can be chosen as u. The key is to recognise the original integral when it reappears.

When the same integral returns, collect those terms together.

Working

Exercise 11F — Integration by Parts

5a. Use integration by parts to find $\int x \cos 4x \, dx$.

5a

5b. Use your answer to part a to find $\int x^2 \sin 4x \, dx$.

5b

6a. Find $\int \sqrt{8-x}\,dx$.

b. Using integration by parts or otherwise, show that

$$\int (x-2)\sqrt{8-x}\,dx = -\frac{2}{5}(8-x)^{3/2}(x+2) + C.$$

c. Hence find $\int_4^7 (x-2)\sqrt{8-x}\,dx$.

6a

6c

6b

7a. Find $\int \sec^2 3x\,dx$.

b. Using integration by parts or otherwise, find $\int x\sec^2 3x\,dx$.

c. Hence show that $\int_{\pi/18}^{\pi/9} x\sec^2 3x\,dx = p\pi - q\ln 3$, finding the exact values of p and q.

7a

7b

7c

Extension

Find $\int \arcsin x \, dx$.

Hint: choose $dv/dx = 1$.

Extension

STEP I 2014, Q2

i. Show that
$$\int \ln(2-x)\,dx$$
$$= -(2-x)\ln(2-x) + (2-x) + C$$
where $x < 2$.

ii. Sketch curve A, given by $y = \ln|x^2 - 4|$.

iii. Show that the area of the finite region enclosed by the positive x-axis, the y-axis and curve A is $4\ln(2+\sqrt{3}) - 2\sqrt{3}$.

iv. Curve B is given by $y = |\ln|x^2 - 4||$. Find the area between B and the x-axis with $|x| < 2$.

Note: you may assume that $t \ln t \to 0$ as $t \to 0^+$.

You will need these standard results from the formula booklet. We will prove them later.

$$\int \tan x \, dx = \ln|\sec x| + C.$$

$$\int \cot x \, dx = \ln|\sin x| + C.$$

$$\int \sec x \, dx = \ln|\sec x + \tan x| + C.$$

$$\int \operatorname{cosec} x \, dx = -\ln|\operatorname{cosec} x + \cot x| + C.$$

Using Partial Fractions 50

Partial fractions split some rational expressions into simpler terms. This lets us integrate expressions with more complicated denominators.

Find $\int \frac{2}{x^2-1}\,dx$.

Working

Further Partial-Fraction Examples 51

Find $\int \frac{x-5}{(x+1)(x-2)}\,dx$.

1

Find $\int \frac{8x^2-19x+1}{(2x+1)(x-2)^2}\,dx$.

2

Test Your Understanding — Partial Fractions 52

Edexcel C4 June 2009, Q3

$$f(x) = \frac{4-2x}{(2x+1)(x+1)(x+3)} = \frac{A}{2x+1} + \frac{B}{x+1} + \frac{C}{x+3}.$$

a. Find A, B, C.

b(i). Hence find $\int f(x)\,dx$.

b(ii). Find $\int_0^2 f(x)\,dx$ in the form $\ln k$.

a

bi

bii

Integrating Improper Algebraic Fractions 53

Find $\int \frac{x^2}{x+1}\,dx$.

How can we simplify a fraction whose numerator has the higher degree?

Simplify

Now integrate.

Integrate

Test Your Understanding — Algebraic Fractions 54

Find $\int \frac{x}{x-1}\,dx$.

1

Find $\int \frac{x^3+2}{x+1}\,dx$.

2

3. $f(x) = \frac{4}{(2x+1)(1-2x)}$, $x \neq \pm\frac{1}{2}$.

a. Given $f(x) = \frac{A}{2x+1} + \frac{B}{1-2x}$, find A and B.

b. Hence find $\int f(x)\,dx$, as a single logarithm plus a constant.

c. Find $\int_1^2 f(x)\,dx$ in the form $\ln k$, with k rational.

3a

3b

3c

4. $f(x) = \frac{17-5x}{(3+2x)(2-x)^2}$, $-\frac{3}{2} < x < 2$.

a. Express $f(x)$ in partial fractions.

b. Hence find $\int_0^1 f(x)\,dx$ in the form $a + \ln b$.

4a

4b

5. $f(x) = \frac{9x^2+4}{9x^2-4}$, $x \neq \pm\frac{2}{3}$.

a. Given $f(x) = A + \frac{B}{3x-2} + \frac{C}{3x+2}$, find A, B, C.

b. Hence find $\int_{-1/3}^{1/3} f(x)\,dx$ in the form $a + b\ln c$, with a, b, c rational.

5a

5b

6. $f(x) = \frac{6+3x-x^2}{x^3+2x^2}$, $x > 0$.

a. Express $f(x)$ in partial fractions.

b. Hence find $\int_2^4 f(x)\,dx$ in the form $a + \ln b$, with a, b rational.

6a

6b

7. $\frac{32x^2+4}{(4x+1)(4x-1)} = A + \frac{B}{4x+1} + \frac{C}{4x-1}$.

a. Find A, B, C.

b. Hence find $\int_1^2 \frac{32x^2+4}{(4x+1)(4x-1)}\,dx$ in the form $2 + k\ln m$, with k, m rational.

7a

7b

Exercise 11G — Extension 60

Extension — MAT 2003, 1D

What is the exact value of

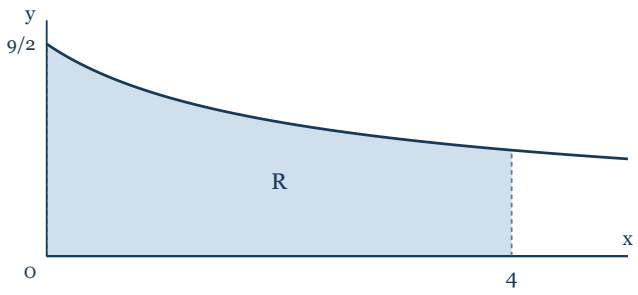
$$\int_1^2 \frac{dx}{x + x^3}?$$

Solution

Finding Areas

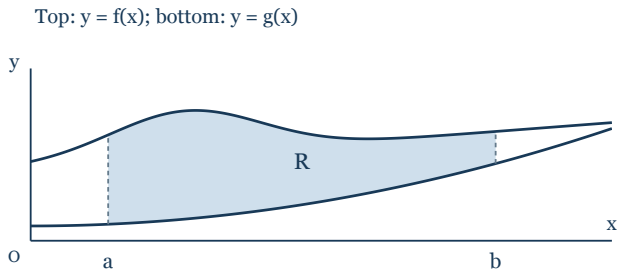
Definite integration gives signed area between a curve and the x-axis; when the curve is above the axis, this is its area. Our new integration skills allow more types of curve.

The curve is $y = 9/\sqrt{4+3x}$. Region R is bounded by the curve, the x-axis and $x = 0$, $x = 4$. Find its area.



Area

Area Between Two Curves



For $f(x) \geq g(x)$ throughout $[a, b]$, the area between the curves is

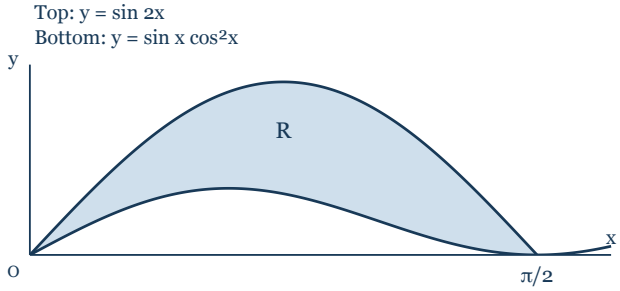
$$R = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$$

$$= \int_a^b [f(x) - g(x)]\,dx.$$

Use top curve minus bottom curve. Split the interval if their order changes.

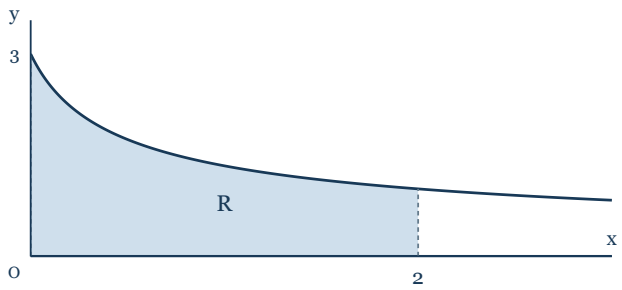
The curves are $y = \sin 2x$ and $y = \sin x \cos^2 x$, for $0 \leq x \leq \pi/2$. Find the area R bounded by them.

Area



Edexcel C4 January 2009, Q2a

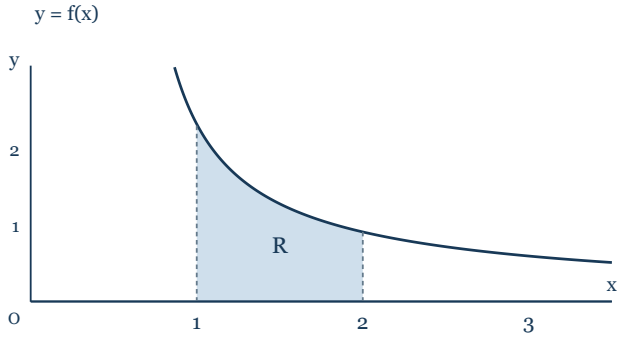
The curve is $y = 3/\sqrt{1+4x}$. Region R is bounded by the curve, the x-axis and $x = 0$, $x = 2$. Use integration to find its area.



Area

3. $f(x) = \frac{4x+3}{(x+2)(2x-1)}$, $x > \frac{1}{2}$.

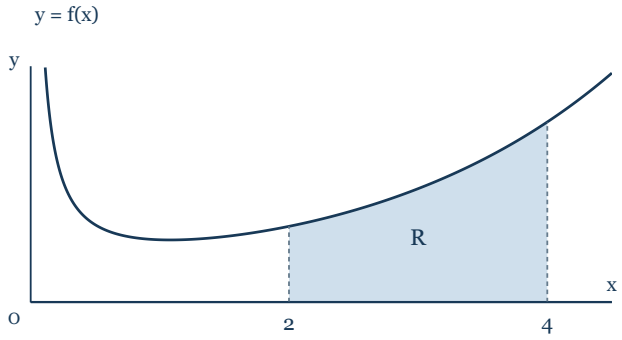
Find the shaded area bounded by $y = f(x)$, the x-axis and $x = 1$, $x = 2$.



3

4. $f(x) = e^{0.5x} + 1/x,\ x > 0.$

Find the shaded area bounded by $y = f(x)$, the x-axis and $x = 2$, $x = 4$.

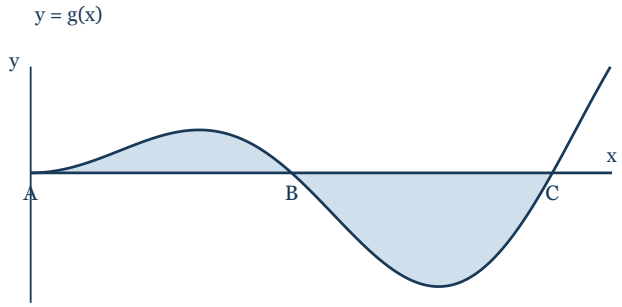


4

5. The curve is $y = g(x) = x \sin x$.

a. Write down the coordinates of A, B, C.

b. Find the total shaded area.



5a

5b

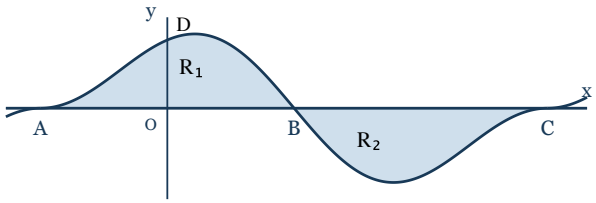
7. The curve is $y = 3\cos x\sqrt{\sin x + 1}$.

a. Find A, B, C, D.

b. Use a suitable substitution to find $\int 3\cos x\sqrt{\sin x + 1}\,dx$.

c. Show the two shaded regions have the same area and express that area as $\sqrt{a}$, finding the positive integer a.

D is the y-intercept; shaded regions R_1 and R_2



7a

7b

7c

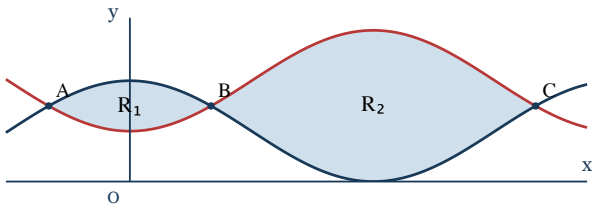
9. The curves are $y = 2\cos x + 2$ and $y = -2\cos x + 4$.

a. Find A, B, C.

b. Find R_1 in the form $a\sqrt{3} + b\pi/c$, with integer a, b, c.

c. Show $R_2 : R_1 = (3\sqrt{3} + 2\pi) : (3\sqrt{3} - \pi)$.

Blue: y = 2 cos x + 2; red: y = −2 cos x + 4

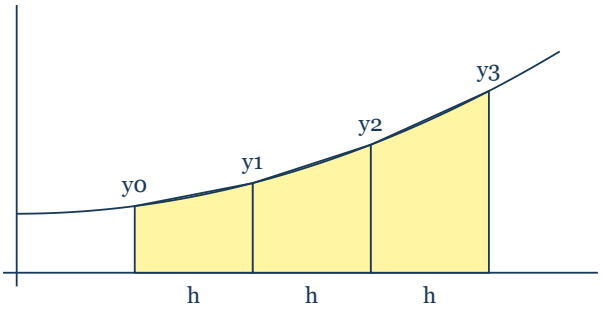


9a

9b

9c

The Trapezium Rule 69



Exact integration can be difficult. Approximate the area under the graph by dividing it into trapeziums of equal width h.

$$\text{Trapezium area} = \frac{h}{2}(y_0 + y_1) + \frac{h}{2}(y_1 + y_2) + \frac{h}{2}(y_2 + y_3)$$

$$= \frac{h}{2}[y_0 + 2(y_1 + y_2) + y_3].$$

Trapezium Rule — Formula and Example 70

In general

$$\int_a^b y\,dx \approx \frac{h}{2}[y_0 + 2(y_1 + \cdots + y_{n-1}) + y_n].$$

The integral represents signed area (area when $y \geq 0$); $\approx$ means approximately equal; $h = (b-a)/n$ is the width of each trapezium.

Approximate the region bounded by $x = 1$, $x = 3$, the x-axis and $y = x^2$, using four strips.

The x-values:

x values

The corresponding y-values:

y values

Width:

Width

Area:

Area

Trapezium Rule — Examination Example 71

Edexcel C2 May 2013 (R), Q2

$y = x/\sqrt{1+x}$.

a. Complete the table at $x = 1.3$, to 4 decimal places.

x	1	1.1	1.2	1.3	1.4	1.5
y	0.7071	0.7591	0.8090		0.9037	0.9487

Calculator tip: on the source Casio model, choose Mode → 3 (Table), use Alpha to enter X, then press =. Menu names vary by model.

a

b. Use the trapezium rule with all table values to estimate $\int_1^{1.5} \frac{x}{\sqrt{1+x}}\,dx$, to 3 decimal places. Show every stage.

b

Further Example — Comparing Trapezium Estimates 72

Given $I = \int_0^{\pi/3} \sec x\,dx$:

a. Use $\int \sec x\,dx = \ln|\sec x + \tan x| + C$ to find I exactly.

b. Estimate I with two trapeziums.

c. Estimate I with four trapeziums.

d. Find the percentage error of each estimate.

a

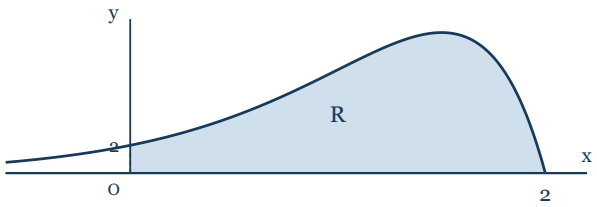
c

b

d

Edexcel C4 June 2014 (R), Q2

$y = (2 - x)e^{2x}$. Region R is bounded by the curve and the coordinate axes.



x	0	0.5	1	1.5	2
y	2	4.077	7.389	10.043	0

a. Use all table values in the trapezium rule to approximate the area, to 2 d.p.

b. Explain how to improve the approximation.

c. Use calculus to find the exact area in simplest form, showing each step.

a

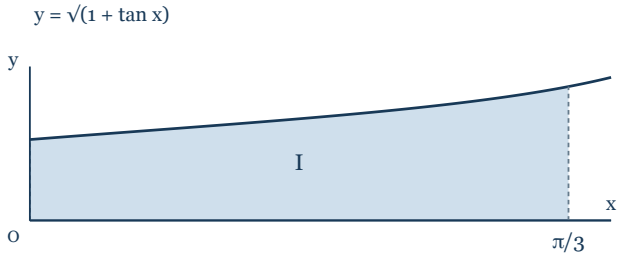
b

c

1. $y = \sqrt{1 + \tan x}$, $0 \leq x \leq \pi/3$.

a. Complete the missing table entries.

x	0	$\pi/12$	$\pi/6$	$\pi/4$	$\pi/3$
y	1		1.2559		1.6529



1a

1b

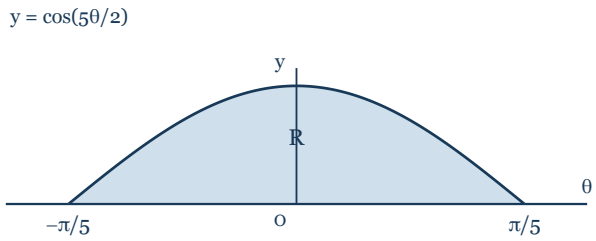
Given $I = \int_0^{\pi/3} \sqrt{1 + \tan x}\, dx$, use the trapezium rule:

b(i). With $x = 0, \pi/6, \pi/3$, estimate I to 4 s.f.

b(ii). With all five table values, estimate I to 4 s.f.

2. Region R lies between $y = \cos(5\theta/2)$ and the horizontal axis, for $-\pi/5 \leq \theta \leq \pi/5$.

θ	$-\pi/5$	$-\pi/10$	0	$\pi/10$	$\pi/5$
y	0		1		0



a. Complete the table to 4 d.p.

b. Use all table values to estimate the area with the trapezium rule, to 3 d.p.

c. State whether this is an underestimate or overestimate, with a reason.

d. Integrate to find the exact area.

e. Calculate the percentage error in part b.

2a

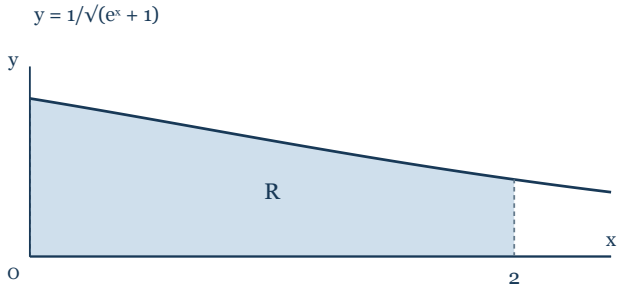
2b

2c

2d

2e

3. $y = 1/\sqrt{e^x + 1}$. Region R is bounded by the curve, both coordinate axes and $x = 2$.



a. Complete the table to 3 d.p.

x	0	0.5	1	1.5	2
y	0.707	0.614	0.519		0.345

b. Use all table values in the trapezium rule to estimate the area of R, to 2 d.p.

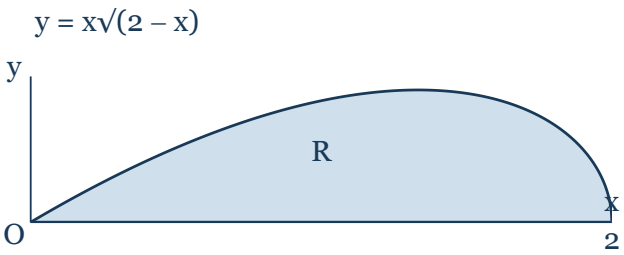
3a

3b

5. $y = x\sqrt{2-x}$, $0 \le x \le 2$.

a. Complete the table at $x = 1.5$.

x	0	0.5	1	1.5	2
y	0	0.6124	1		0



Given $I = \int_0^2 x\sqrt{2-x}\,dx$:

b. Use four trapeziums to estimate I, to 4 s.f.

c. Use a substitution or otherwise to find I exactly in the form $2^q p$, for rational p, q.

d. Calculate the percentage error of part b.

5a

5b

5c

5d

7. $I = \int_0^3 e^{\sqrt{2x+1}}\,dx$.

a. Complete the table for $y = e^{\sqrt{2x+1}}$.

x	0	0.5	1	1.5	2	2.5	3
y	2.7183				9.3565	11.5824	14.0940

b. Use all table values in the trapezium rule to estimate I, to 4 s.f.

c. Use $t = \sqrt{2x+1}$ to express I as $\int_a^b kte^t\,dt$, finding a, b, k.

d. Integrate by parts and find I, to 4 s.f.

7a

7b

7c

7d

Integration with Parametric Equations 79

Suppose $x = t^2$ and $y = t + 1$. To find area we need $\int y\,dx$, but y is expressed in terms of t. Rewrite the integral:

$$\int y\,dx = \int y\frac{dx}{dt}\,dt.$$

Memory tip: $dx = (dx/dt)\,dt$, the substitution rule derived from the chain rule. Informally the dt factors appear to cancel.

Find the area bounded by $x = t^2$, $y = t + 1$, $t > 0$, the x-axis and $x = 0$, $x = 3$.

Step 1: find dx/dt.

Step 2: change the limits to t-values.

Step 3: integrate.

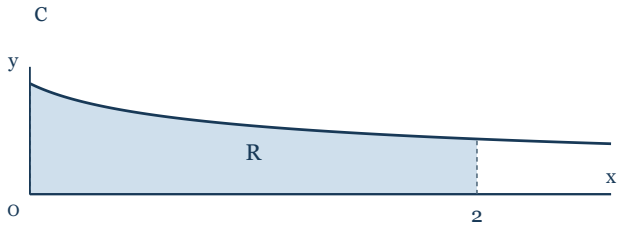
Derivative

Limits

Integrate

Further Parametric Area Example 80

Curve C has parametric equations $x = t(1 + t)$, $y = 1/(1 + t)$, $t \geq 0$. Find the exact area R, bounded by C, the x-axis and $x = 0$, $x = 2$.



Derivative

Lower limit

Upper limit

Integral

Edexcel C4 January 2013, Q5

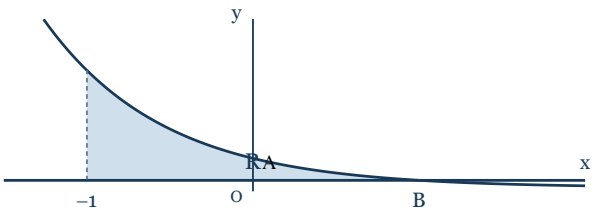
Curve C: $x = 1 - \frac{1}{2}t$, $y = 2^t - 1$. It crosses the y-axis at A and the x-axis at B.

a. Show that $A = (0, 3)$.

b. Find the x-coordinate of B.

d. Find the exact area R, bounded by C, $x = -1$ and the x-axis.

C; A is the y-intercept



Tip: $\frac{d}{dx}a^x = a^x \ln a$, and $\int a^x \, dx = a^x / \ln a + C$ for $a > 0$, $a \neq 1$.

a

b

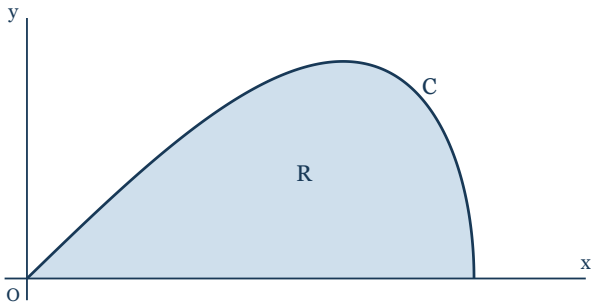
d

1. Curve C has parametric equations $x = t^3$, $y = t^2$, $t \geq 0$. Show that the exact area bounded by the curve, the x-axis and $x = 0$, $x = 4$ is $k\sqrt[3]{2}$, where k is a rational constant to be found.

1

Parametric Area — Question 2 (source slide 82) 83

2. Curve C has parametric equations $x = \sin t,\ y = \sin 2t,\ 0 \le t \le \pi/2$. The finite region R is bounded by the curve and the x-axis. Find its exact area. (6 marks)



2

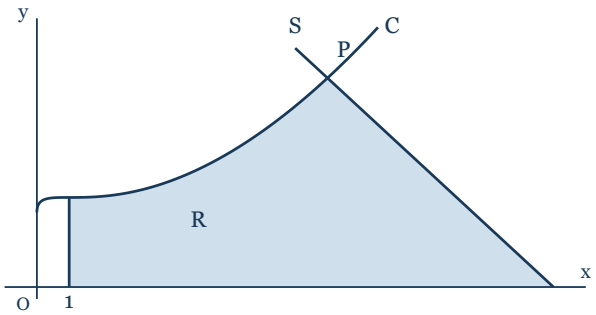
Parametric Area — Question 3 (source slide 82) 84

3. Curve C has parametric equations $x = (t+1)^2,\ y = \frac{1}{2}t^3 + 3,\ t \ge -1$. Point P has $t = 2$, and S is the normal at P.

(a) Find an equation of S. (5 marks)

Region R is bounded by C, S, the x-axis and $x = 1$.

(b) Use integration to find its area. (5 marks)



3a

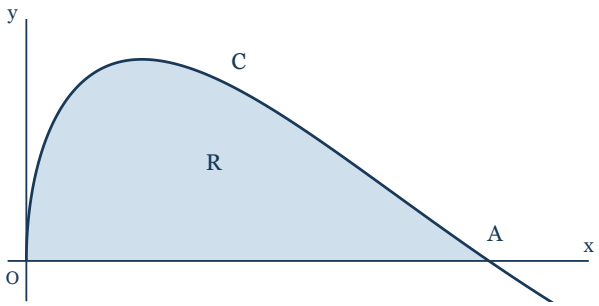
3b

Parametric Area — Question 4 (source slide 82) 85

4. Curve C has parametric equations $x = 3t^2,\ y = \sin 2t,\ t \geq 0$.

(a) State t at point A, the first positive crossing of the x-axis. (1 mark)

(b) Find, in terms of π, the exact shaded area bounded by C and the x-axis. (6 marks)



4a

4b

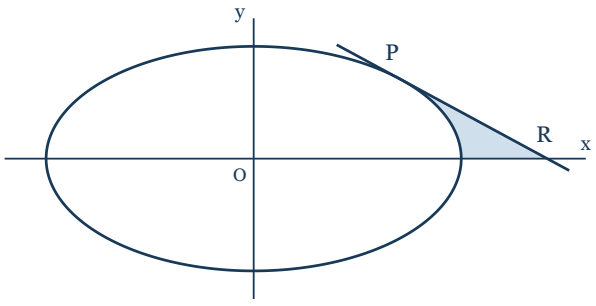
Parametric Area — Question 5 (source slide 82) 86

5. Curve C has parametric equations $x = 5\cos\theta,\ y = 4\sin\theta,\ 0 \leq \theta < 2\pi$. At $P, \theta = \pi/4$.

(a) Find the gradient at P. (3 marks)

(b) Find the tangent equation at P. (3 marks)

(c) Find the exact shaded area bounded by tangent PR, the curve and the x-axis. (6 marks)



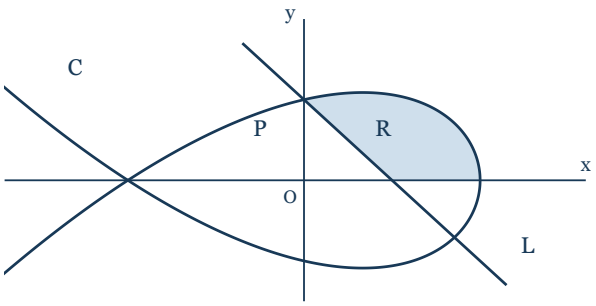
5a

5b

5c

Parametric Area — Question 6 (source slide 82) 87

6. Curve C has parametric equations $x = 1 - t^2,\ y = 2t - t^3,\ t \in \mathbb{R}$. Line L is the normal at P, where C intersects the positive y-axis. Find the exact area of R, bounded by C, L and the x-axis, as shown. (7 marks)



6

Parametric Area — Question 7 (source slide 82) 88

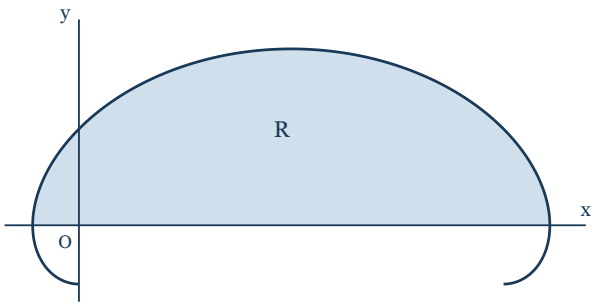
7. Curve C has parametric equations $x = t - 2\sin t,\ y = 1 - 2\cos t,\ 0 \le t \le 2\pi$.

(a) Show it crosses the x-axis at $t = \pi/3, 5\pi/3$. (3 marks)

Region R is enclosed by C and the x-axis.

(b) Show its area is $\int_{\pi/3}^{5\pi/3} (1 - 2\cos t)^2\, dt$. (3 marks)

(c) Evaluate this integral exactly. (4 marks)



7a

7b

7c

Differential Equations — Separating Variables 89

Differential equations contain variables and derivatives, for example x, y and dy/dx. Solving means expressing the relationship between the variables without a derivative remaining.

Find the general solution of $dy/dx = xy + y$.

Step 1

Factorise if needed, then separate: put the y-terms with dy/dx, and the x-terms on the other side.

Separate

Step 2

Integrate. The chain rule lets $(dy/dx)\,dx$ become dy.

Integrate

Step 3

Make y the subject if asked.

Subject

Tip: combine $e^{f(x)+C}$ as $e^{C}e^{f(x)}$. The factor e^{C} is positive; a sign can be absorbed into a new constant when undoing $\ln|y|$.

Another Separable Differential Equation 90

Find the general solution of $(1+x^2)\frac{dy}{dx} = x\tan y$.

Step 1

Separate the variables.

Separate

Step 2

Integrate.

Integrate

Step 2b

Combine the constant with the logarithms.

Constant

Step 3

Make y the subject on a chosen branch.

Subject

Differential Equations with a Given Condition 91

Solve $\frac{dy}{dx} = -\frac{3(y-2)}{(2x+1)(x+2)}$, given $y = 4$ when $x = 1$. Leave the answer as $y = f(x)$.

Working

Test Your Understanding — Differential Equations 92

Edexcel C4 January 2012, Q4

Given $y = 2$ at $x = \pi/4$, solve

$$\frac{dy}{dx} = \frac{3}{y\cos^2 x}.$$

Working

Key Points on Differential Equations

1. Separate variables: put the y-terms on the left by dividing, factorising first if necessary. Check whether dividing loses constant solutions.
2. When logarithms occur, it can help to write the integration constant as $\ln k$, with $k > 0$.
3. Combine logarithms using their laws. For example:

$$2\ln|x+1| - \ln|x| \quad \longrightarrow$$

$$\ln\left|\frac{(x+1)^2}{x}\right|.$$

4. Substitute the given conditions to determine the constant; it is often easier to do this early.
5. Examination questions often combine partial fractions with differential equations.

Exercise 11J — Pages 324–326

4. Given $x = 0$ when $y = 0$, find a particular solution of $(2y + 2yx)\frac{dy}{dx} = 1 - y^2$, in the form $y = g(x)$.

4

5. Find the general solution of $e^{x+y}\frac{dy}{dx} = 2x + xe^y$, in the form $\ln|g(y)| = f(x)$.

5

6. Solve $(1-x^2)\frac{dy}{dx} = xy + y$, given $y = 6$ at $x = 0.5$. Give $y = f(x)$.

6

7. Solve $(1+x^2)\frac{dy}{dx} = x - xy^2$, given $y = 2$ at $x = 0$. Give $y = f(x)$.

7

8. Solve $dy/dx = xe^{-y}$, given $y = \ln 2$ at $x = 4$. Give $y = f(x)$.

8

9. Solve $dy/dx = \cos^2 y + \cos 2x \cos^2 y$, given $y = \pi/4$ at $x = \pi/4$. Give $\tan y = f(x)$.

9

10. Given $y = 1$ at $x = \pi/2$, solve $dy/dx = xy \sin x$.

10

11a. Find $\int \frac{3x+4}{x}\,dx$, $x > 0$.

11a

11b. Given $y = 16$ at $x = 1$, solve

$$\frac{dy}{dx} = \frac{3x\sqrt{y} + 4\sqrt{y}}{x},$$

giving $y = f(x)$.

11b

12a. Express $\frac{8x-18}{(3x-8)(x-2)}$ in partial fractions.

b. For $x \geq 3$, find the general solution of

$$(x-2)(3x-8)\frac{dy}{dx} = (8x-18)y.$$

c. Hence find the particular solution with $y = 8$ at $x = 3$, as $y = f(x)$.

12a

12b

12c

14a. Find the general solution of $dy/dx = -1/(x+2)^2$.

b. On the same axes, sketch three particular solutions.

c. Write down the solution through $(8, 3.1)$.

14a

14b

14c

14b explanation

15a. Show that the general solution of $dy/dx = -x/y$ can be written $x^2 + y^2 = C$.

b. Sketch three particular solution curves on the same axes.

c. Write down the solution through $(0, 7)$.

15a

15b

15c

15b explanation

Extension — STEP I 2013, Q7

i. Use $y = ux$, where u is a function of x, to show that the solution of

$$\frac{dy}{dx} = \frac{x}{y} + \frac{y}{x}, \qquad x > 0,\ y > 0,$$

with $y = 2$ when $x = 1$, is

$$y = x\sqrt{4 + 2\ln x}, \qquad x > e^{-2}.$$

ii. Use a substitution to solve

$$\frac{dy}{dx} = \frac{x}{y} + \frac{2y}{x}, \qquad x > 0,\ y > 0,$$

with $y = 2$ at $x = 1$.

iii. Solve

$$\frac{dy}{dx} = \frac{x^2}{y} + \frac{2y}{x}, \qquad x > 0,\ y > 0,$$

with $y = 2$ at $x = 1$.

Solutions

In real-life models, a rate of change can depend on the current value. This extends the exponential growth studied in Year 1.

The population P of rabbits increases at a rate proportional to P. Form and solve a differential equation, with time t.

Form equation

Solve

Forming Differential Equations — Draining a Tank 103

A cylindrical tank has diameter 20 m. Water flows out at a rate proportional to the cube root of its volume V. Time t is in minutes and depth h is in metres.

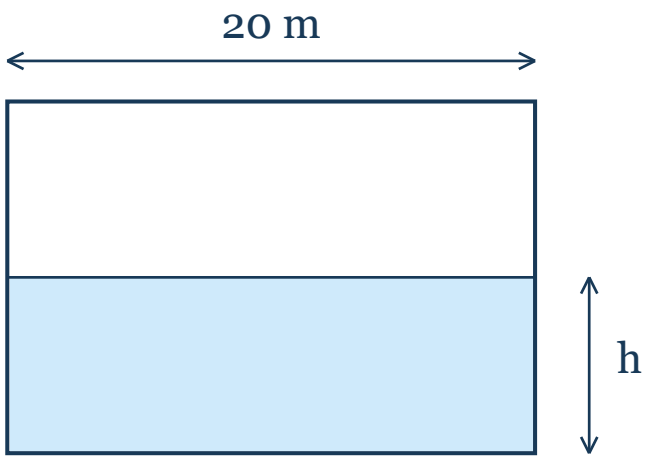
a. Show that $dh/dt = -k\sqrt[3]{h}$.

b. Find a general solution $h = (P - Qt)^{3/2}$.

Initially $h = 27$; after 10 minutes, $h = 8$.

c. Find P, Q.

d. Find the time when $h = 1$.



a

b

c

d

Test Your Understanding — Inflow and Outflow 104

Edexcel C4 June 2005, Q8

Liquid enters an initially empty container at $20\ \text{cm}^3/\text{s}$. It leaves at a rate proportional to the volume V.

a. Explain why $dV/dt = 20 - kV$, with $k > 0$.

b. Solve to show $V = A + Be^{-kt}$, giving A, B in terms of k.

c. When $t = 5$, $dV/dt = 10$. Find V when $t = 10$.

a

b

c

1. A rabbit population P satisfies $dP/dt = kP$, with $k > 0$ and time t in years. Initially there are 200 rabbits.

a. Find P in terms of k, t.

b. For $k = 3$, find when the population reaches 4000.

c. Explain a limitation of the model for large t.

1a

1b

1c

2. The mass M of a plant's leaves varies with time according to $\dfrac{dM}{dt} = M - M^2$.

a. Given $M = 0.5$ at $t = 0$, find M in terms of t.

b. Find M when $t = \ln 2$.

c. Explain what happens to M as t increases.

Form equation

2a

2b

2c

3. Ice on a pond has thickness x cm, increasing at a rate inversely proportional to x^2. Initially $x = 1$; after 20 days, $x = 2$.

a. Show that $x = \sqrt[3]{\frac{7}{20}t + 1}$.

b. Find the time for the thickness to increase from 2 cm to 3 cm.

3a

3b

4. Tea at temperature $T\,^\circ\text{C}$ cools in a room at $25\,^\circ\text{C}$. Its rate of cooling is proportional to the difference between the tea and room temperatures.

a. Show that $dT/dt = -k(T - 25)$, explaining why $k > 0$.

Initially $T = 85$; 10 minutes later, $T = 55$.

b. Find the temperature after 15 minutes, to 1 decimal place.

4a

4b

5. The surface area A mm^2 of an oil drop, at time t minutes, satisfies

$$\frac{dA}{dt} = \frac{A^{3/2}}{10t^2}.$$

At $t = 1$, $A = 1$.

a. Find A in terms of t.

b. Show that the area cannot exceed $400/361$ mm^2.

5a

5b

6. A bath is modelled as a cuboid with base area 6000 cm^2. Water enters at 12000 cm^3/min and leaves through the plughole at $500h$ cm^3/min, where h is its depth in cm.

a. Show that $60\,dh/dt = 120 - 5h$, with time t in minutes.

Initially $h = 6$.

b. Find t when $h = 10$.

6a

6b

7a. Express $1/[P(10000-P)]$ in partial fractions.

A deer population P satisfies

$$\frac{dP}{dt} = \frac{1}{200}P(10000-P),$$

where t is in years. Initially $P = 2500$.

b. Solve, giving $P = a/(b+ce^{-50t})$.

c. Find the maximum population according to the model.

7a

7c

7b

8. Liquid enters a container at $40\ \text{cm}^3/\text{s}$ and leaks at $\frac{1}{4}V\ \text{cm}^3/\text{s}$, where $V\ \text{cm}^3$ is its volume.

a. Show that $-4\,dV/dt = V - 160$.

Initially $V = 5000$.

b. Find $V = a + be^{-t/4}$, determining a, b.

c. Write down the limiting value of V as $t \to \infty$.

8a

8c

8b

9. In carbon dating, the amount R of radioactive carbon remaining in a fossil decreases at a rate proportional to R. Time t is in years.

a. Initially $R = R_0$. Show that $R = R_0 e^{-kt}$.

The half-life is 5730 years: after this time, half remains.

b. Find k exactly.

c. A fossil has 10% of its expected original carbon remaining. Find its age to the nearest year.

9a

9b

9c

Extension

Try STEP I 2011, Question 7(i) and (ii).

Summary of Functions 114

Recall the method and integral. "Booklet" refers to the formula booklet used by the supplied course.

1. $\int \sin x \, dx$

Standard result.
$-\cos x + C$.
Booklet: No.

2. $\int \cos x \, dx$

Standard result.
$\sin x + C$.
Booklet: No.

3. $\int \tan x \, dx$

Use $\sin x / \cos x$, of the form $kf'(x)/f(x)$.
$\ln|\sec x| + C$.
Booklet: Yes.

4. $\int \sin^2 x \, dx$

Use $\cos 2x = 1 - 2\sin^2 x$:
$\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$.
$\frac{1}{2}x - \frac{1}{4}\sin 2x + C$.
Booklet: No.

5. $\int \cos^2 x \, dx$

Use $\cos 2x = 2\cos^2 x - 1$:
$\cos^2 x = \frac{1}{2} + \frac{1}{2}\cos 2x$.
$\frac{1}{2}x + \frac{1}{4}\sin 2x + C$.
Booklet: No.

6. $\int \tan^2 x \, dx$

Use $1 + \tan^2 x = \sec^2 x$, so $\tan^2 x = \sec^2 x - 1$.
$\tan x - x + C$.
Booklet: No.

7. $\int \operatorname{cosec} x \, dx$

Substitute $u = \operatorname{cosec} x + \cot x$. Derivation beyond this course's exam expectations.
$-\ln|\operatorname{cosec} x + \cot x| + C$.
Booklet: Yes.

8. $\int \sec x \, dx$

Substitute $u = \sec x + \tan x$. Derivation beyond this course's exam expectations.
$\ln|\sec x + \tan x| + C$.
Booklet: Yes.

9. $\int \cot x \, dx$

Use $\cos x / \sin x$, of the form $f'(x)/f(x)$.
$\ln|\sin x| + C$.
Booklet: Yes.

Summary of Functions 115

Recall the method and integral. "Booklet" refers to the formula booklet used by the supplied course.

1. $\int \operatorname{cosec}^2 x \, dx$

By observation.
$-\cot x + C$.
Booklet: No.

2. $\int \sec^2 x \, dx$

By observation.
$\tan x + C$.
Booklet: Yes, but memorise.

3. $\int \cot^2 x \, dx$

Use $1 + \cot^2 x = \operatorname{cosec}^2 x$.
$-\cot x - x + C$.
Booklet: No.

4. $\int \sin 2x \cos 2x \, dx$

For sine and cosine with the same argument, use the double-angle identity:
$\sin 2x \cos 2x = \frac{1}{2} \sin 4x$.
$-\frac{1}{8} \cos 4x + C$.
Booklet: No.

5. $\int 1/x \, dx$

Standard result.
$\ln |x| + C$.
Booklet: No.

6. $\int \ln x \, dx$

Integration by parts: $u = \ln x$, $dv/dx = 1$.
$x \ln x - x + C$.
Booklet: No.

7. $\int \frac{x}{x+1} \, dx$

Algebraic division:
$x/(x+1) = 1 - 1/(x+1)$.
$x - \ln |x+1| + C$.
Booklet: No.

8. $\int \frac{1}{x(x+1)} \, dx$

Partial fractions.
$\ln |x| - \ln |x+1| + C$.
Booklet: No.

Summary of Functions 116

1. $\int \frac{4x}{x^2+1} \, dx$

Reverse chain rule: $kf'(x)/f(x)$.
$2 \ln |x^2 + 1| + C$.

2. $\int \frac{x}{(x^2+1)^2} \, dx$

Consider $y = (x^2+1)^{-1}$ and differentiate, or substitute $u = x^2 + 1$.
$-\frac{1}{2}(x^2+1)^{-1} + C$.

3. Find each integral:

$$\int e^{2x+1} \, dx, \qquad \int \frac{1}{1-3x} \, dx.$$

For a linear inner function, divide by the coefficient of x.

$$\begin{matrix} \frac{1}{2} e^{2x+1} + C \\ -\frac{1}{3} \ln |1 - 3x| + C \end{matrix}.$$

4. $\int x\sqrt{2x+1} \, dx$

Use integration by parts or a sensible substitution: $u = 2x + 1$, or $u^2 = 2x + 1$.
$\frac{1}{15}(2x+1)^{3/2}(3x-1) + C$.

5. $\int \sin^5 x \cos x \, dx$

Reverse chain rule.
$\frac{1}{6} \sin^6 x + C$.